ITAI 2277 AI Applications Portfolio - SPRING 2026

Nicole Marcial

Biomedical AI Data Pipelines | Feature Engineering | Model Workflow Design

Portfolio focused on applying AI concepts to real biomedical data, with hands-on work in DrugBank preprocessing, feature engineering, EDA explanation, and AI project documentation for a group drug-discovery capstone.

Course Highlight 1: CompoundIQ Capstone

CompoundIQ was a team AI capstone project for safe 3-drug combination discovery.

Project Purpose

- Designed a pipeline to evaluate and rank safer 3-drug combinations.
- Connected NLP, molecular generation, engineered drug features, and graph-based prediction.
- Framed as a research-support tool, not medical advice.

My Main Contribution

- Built and explained the DrugBank data engineering component.
- Transformed raw drug records into structured feature tables.
- Created interaction, CYP, and safety-complexity features for downstream modeling.
- Organized the group GitHub repository so the project was easier to review and submit.

Tools Practiced

- Python, pandas, NumPy, lxml, RDKit, NetworkX, scikit-learn.
- Parquet storage for efficient ML-ready files.
- GitHub documentation and technical presentation workflow.

Course Highlight 2: DrugBank Pipeline + EDA

Turning noisy biomedical data into model-ready features

Raw DrugBank XML
large biomedical source

→

Structured Tables
drugs, interactions, targets, enzymes

→

Feature Engineering
interaction counts, CYP features, safety score

→

EDA Insights
sparsity + long-tail behavior

→

Model Support
features for GNN safety scoring

What I Learned

A strong AI model depends on well-prepared data. This project taught me how domain-specific features can make raw biomedical records more useful for downstream machine learning.

What I Practiced

Parsing, deduplication, feature creation, split design, and visualization. Explaining technical results in beginner-friendly language.

Portfolio Value

Shows employer-ready data engineering and AI pipeline thinking. Demonstrates clear communication of model inputs and outputs.

Real-World Applications

How I can apply these skills beyond the classroom

Industries and Teams

- Pharmaceutical AI and drug discovery
- Healthcare analytics and safety screening
- Academic and research data engineering
- AI education tools and technical documentation

Problems I Can Help Tackle

- Clean and structure messy scientific datasets
- Build features that make model predictions more interpretable
- Create reproducible Colab/GitHub workflows
- Present technical AI work to non-expert audiences

Responsible AI: research support, not medical advice

Wrap-Up

Portfolio repository and contact information

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GitHub: https://github.com/2002nicole/ITAI2277-Capstone-Portfolio

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What this portfolio shows:

Individual DrugBank feature engineering work

Group capstone context clearly labeled

Phase 2 presentation included because it reflects my DrugBank work

Phase summaries included for the rest of the group project